

ASTR 302 Fall 2026:
Characteristics of Spectrographs (cont)

Agenda – October 12, 2026

- Class Logistics (Observing, Skill Building, DS9)
- Dispersive Elements Continued
- Spectrographs (Resolution and Signal)

Recommended Reading: Birney Chapter 12

Resolution Simulator: https://andor.oxinst.com/learning/view/article/slit-width-pixel-size-and-spectral-resolution

Characteristics of Spectrographs

Dispersion Power – How much different wavelengths are spread out. This is determined by whatever dispersive element(s) is used.

Resolution – How well your instrument can measure wavelength differences. This is determined by all your optics and detector together. This is usually what an observatory or instrument manual would report.

Resolving Power / Resolution

Resolution (R) – How well your instrument can measure wavelength differences. This is determined by all your optics and detector together. This is usually what an observatory or instrument manual would report.

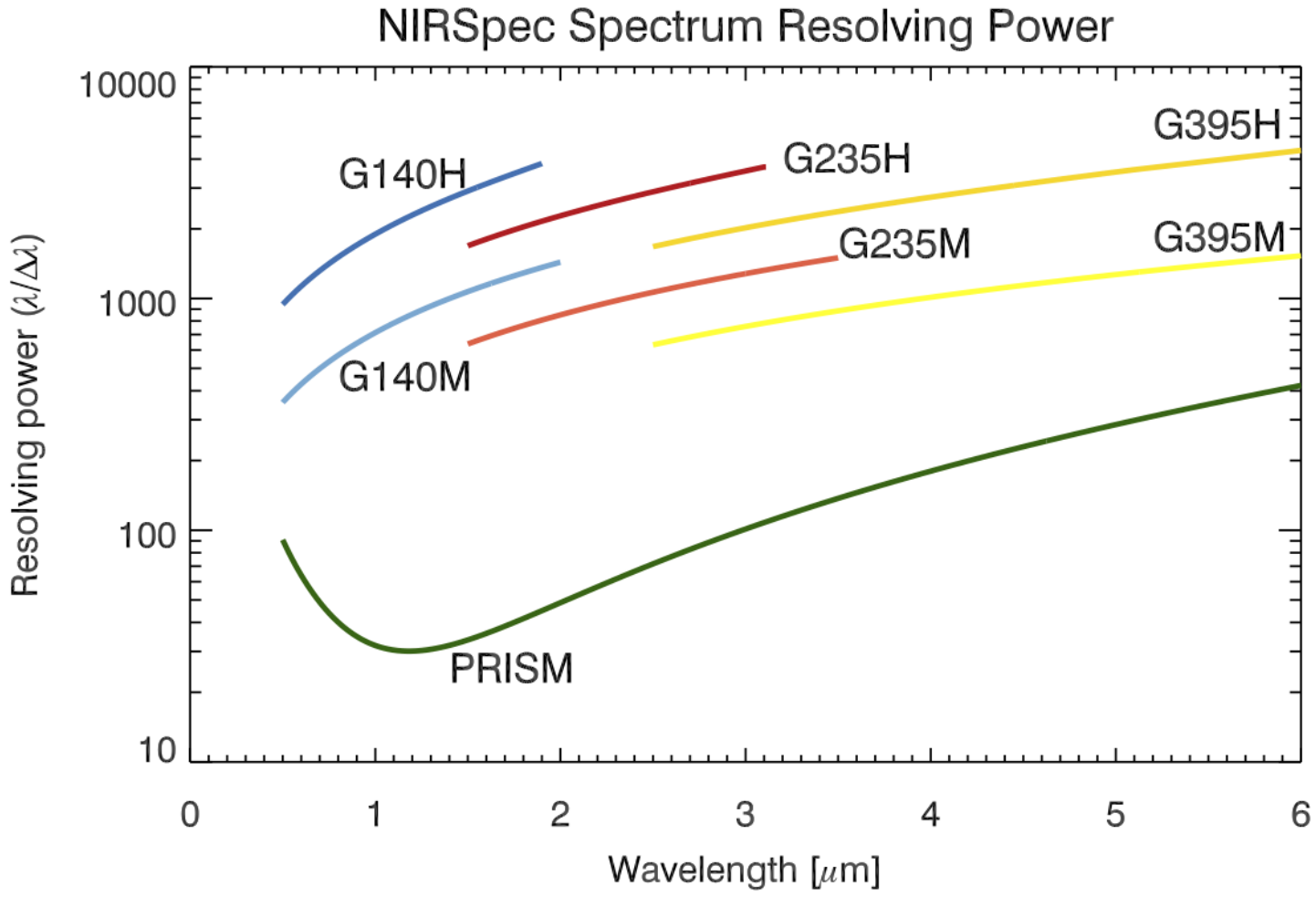
$$R = \frac{\lambda}{\Delta\lambda}$$

$\Delta\lambda$ is the closest wavelength spacing two spectral features can have while still being separated on your detector. λ is the reference wavelength where resolution is estimated.

Resolving Power / Resolution

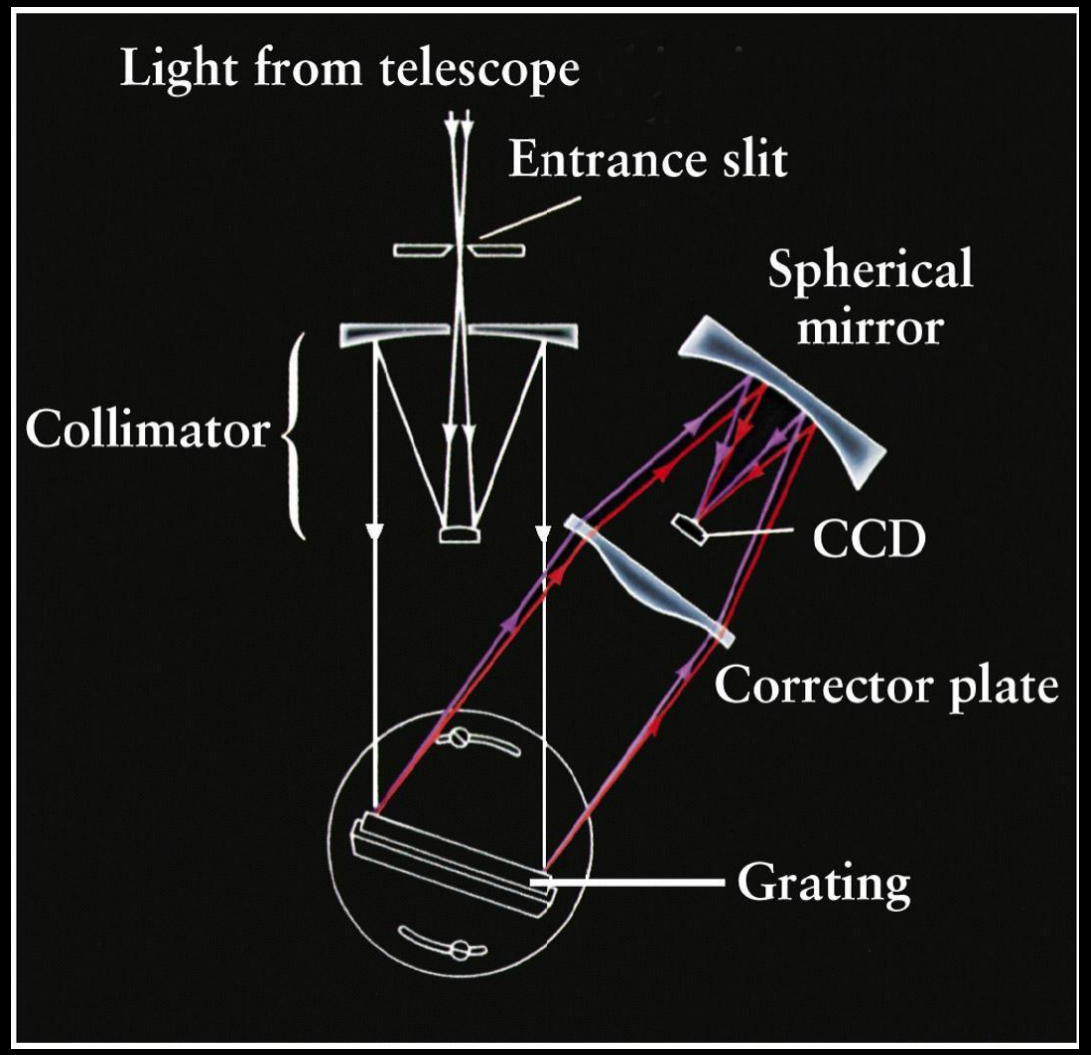
Resolution (R) – is often reported in approximate terms (R ~ 1000), because resolution changes with wavelength. This is because refraction and diffraction are wavelength dependent phenomena.

$$R = \frac{\lambda}{\Delta\lambda}$$



JWST/NIRSpec Resolution for each VPH grating or prism

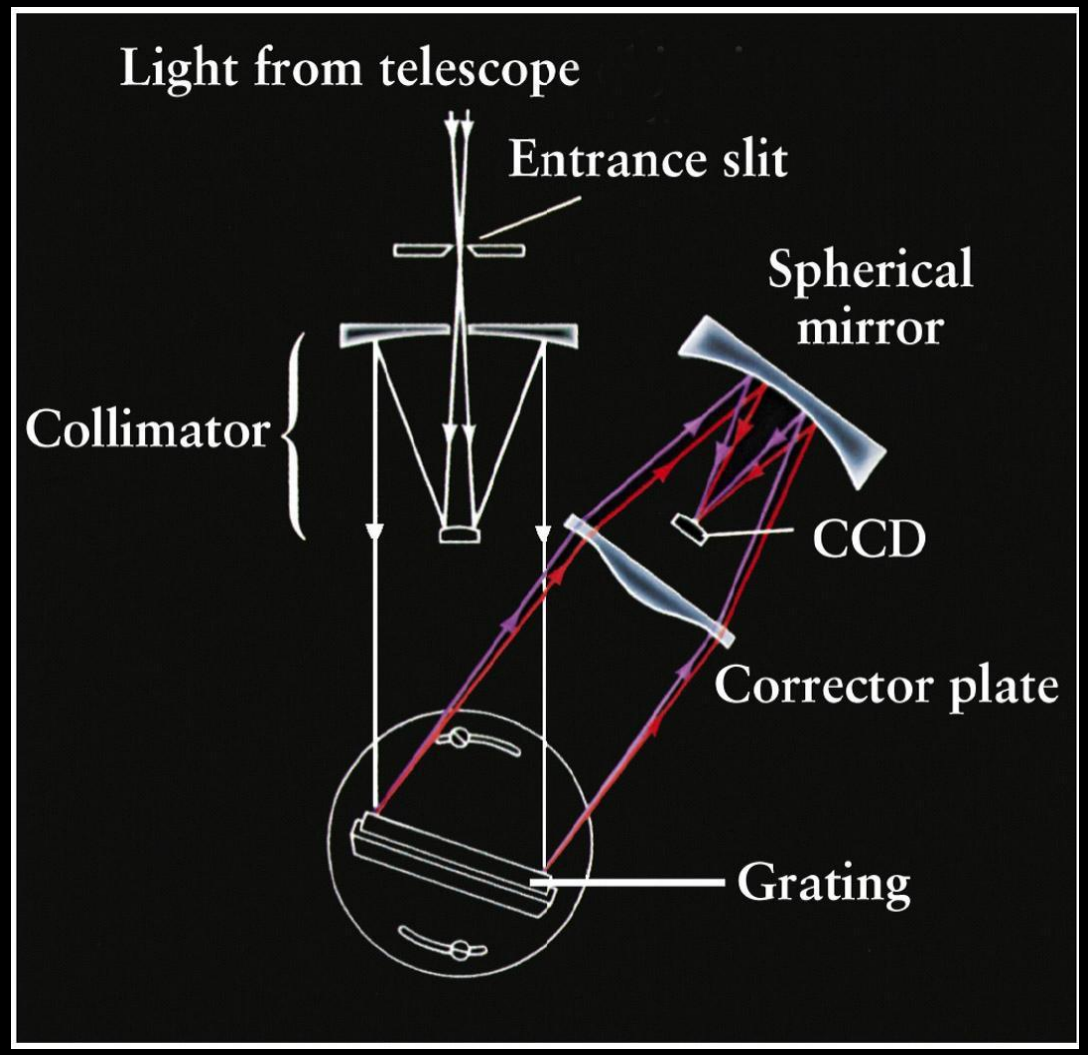
Components of a Spectrograph with Grating

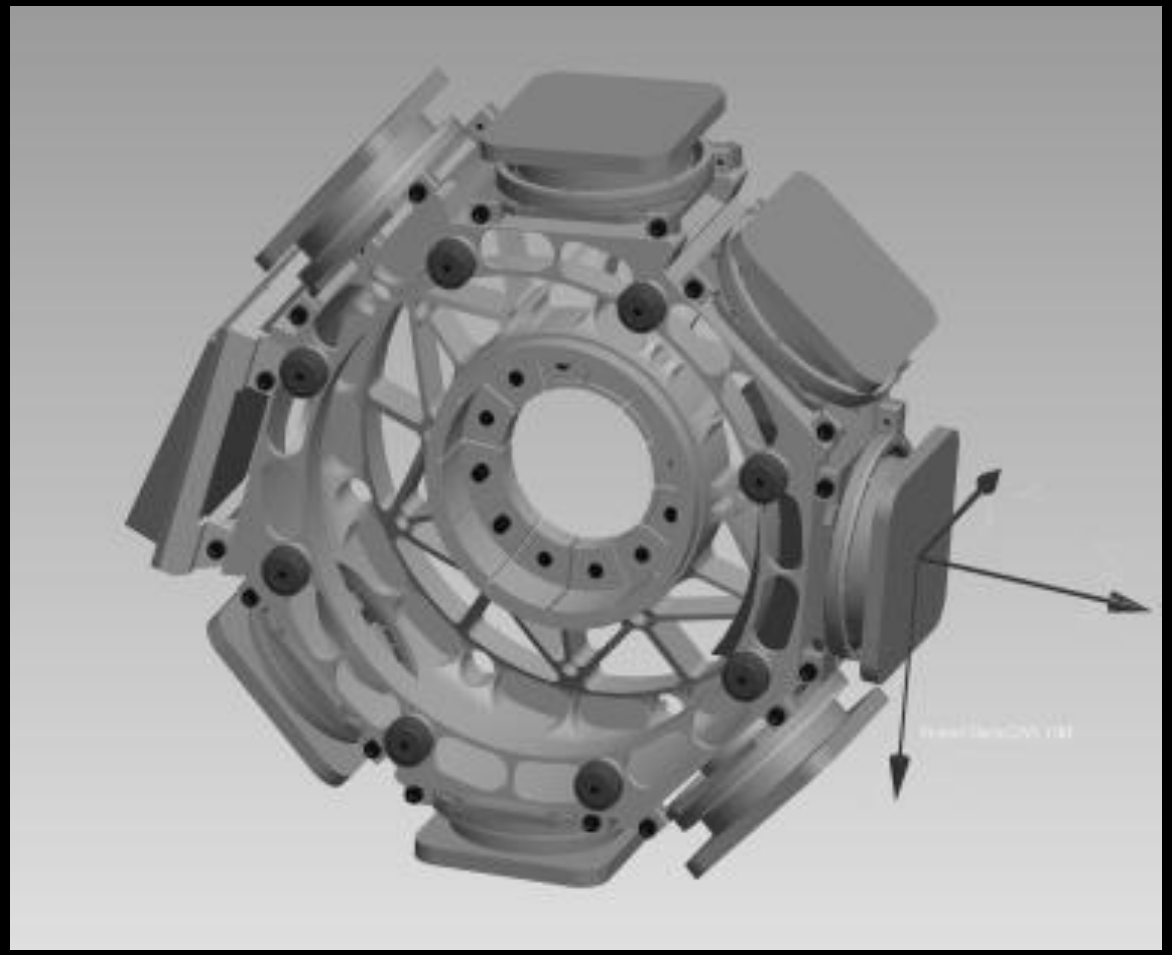


A reflective grating can be rotated to access different orders and wavelengths with your detector.

Some instruments keep gratings fixed, but rotate through several to access different wavelengths.

Components of a Spectrograph with Grating

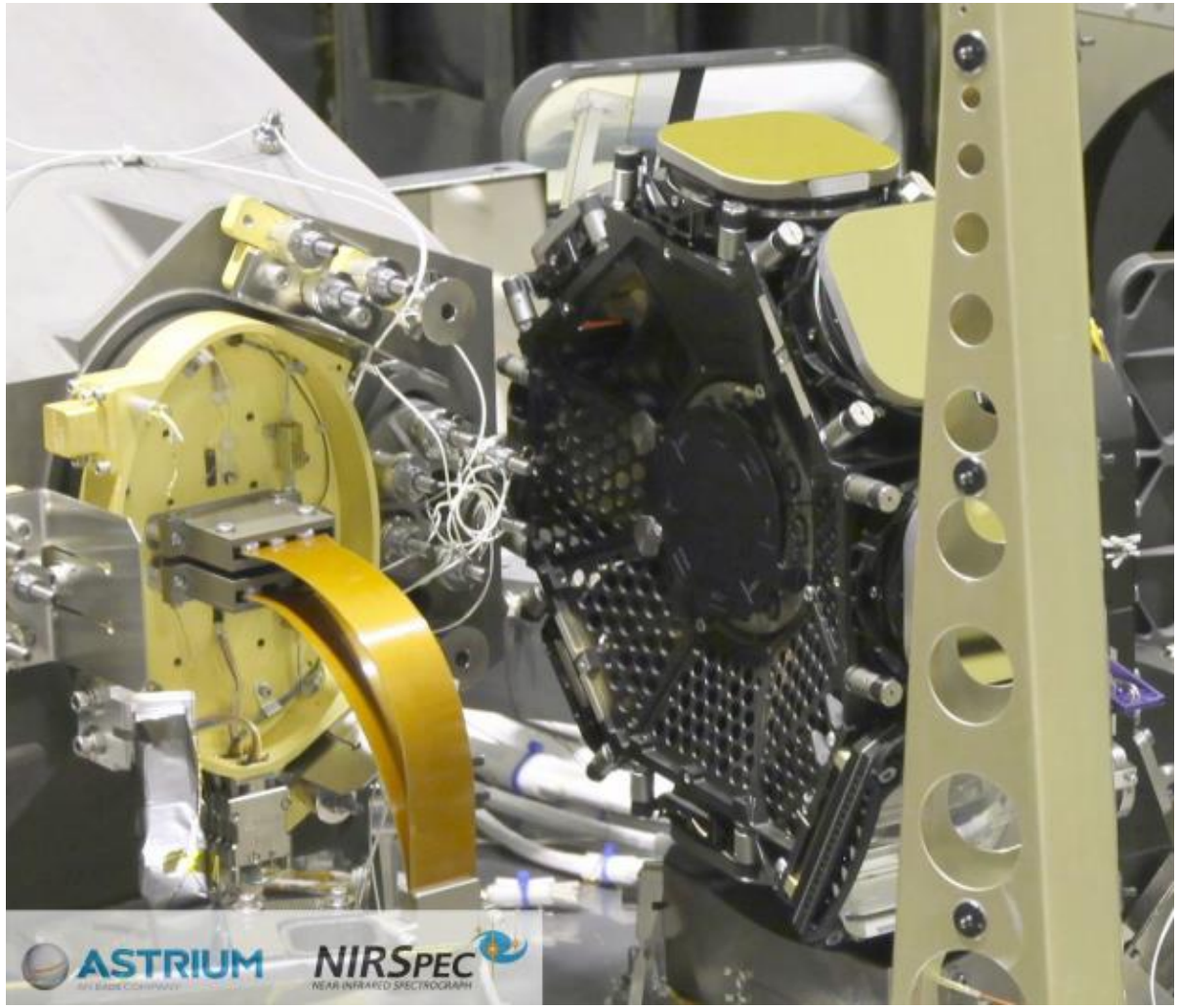


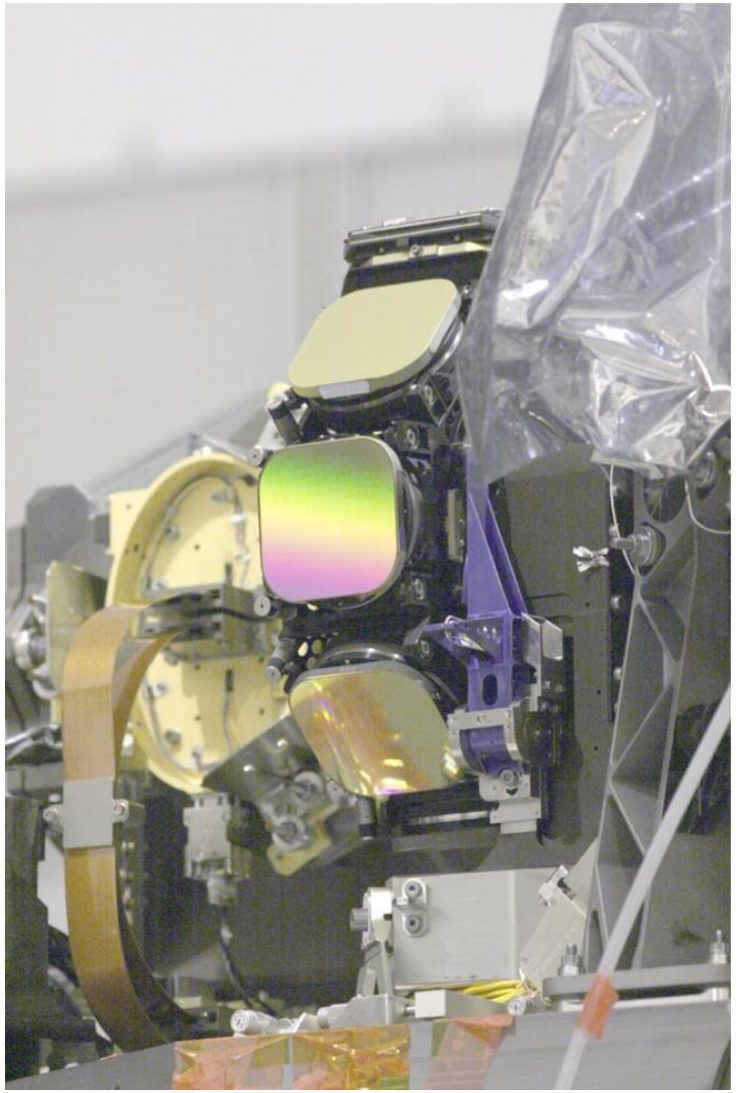


JWST/NIRSpec Grating Wheel Design (Weidlich + 2006)

Or instruments with grating wheels (Keck/Liger, Keck/SCALES)

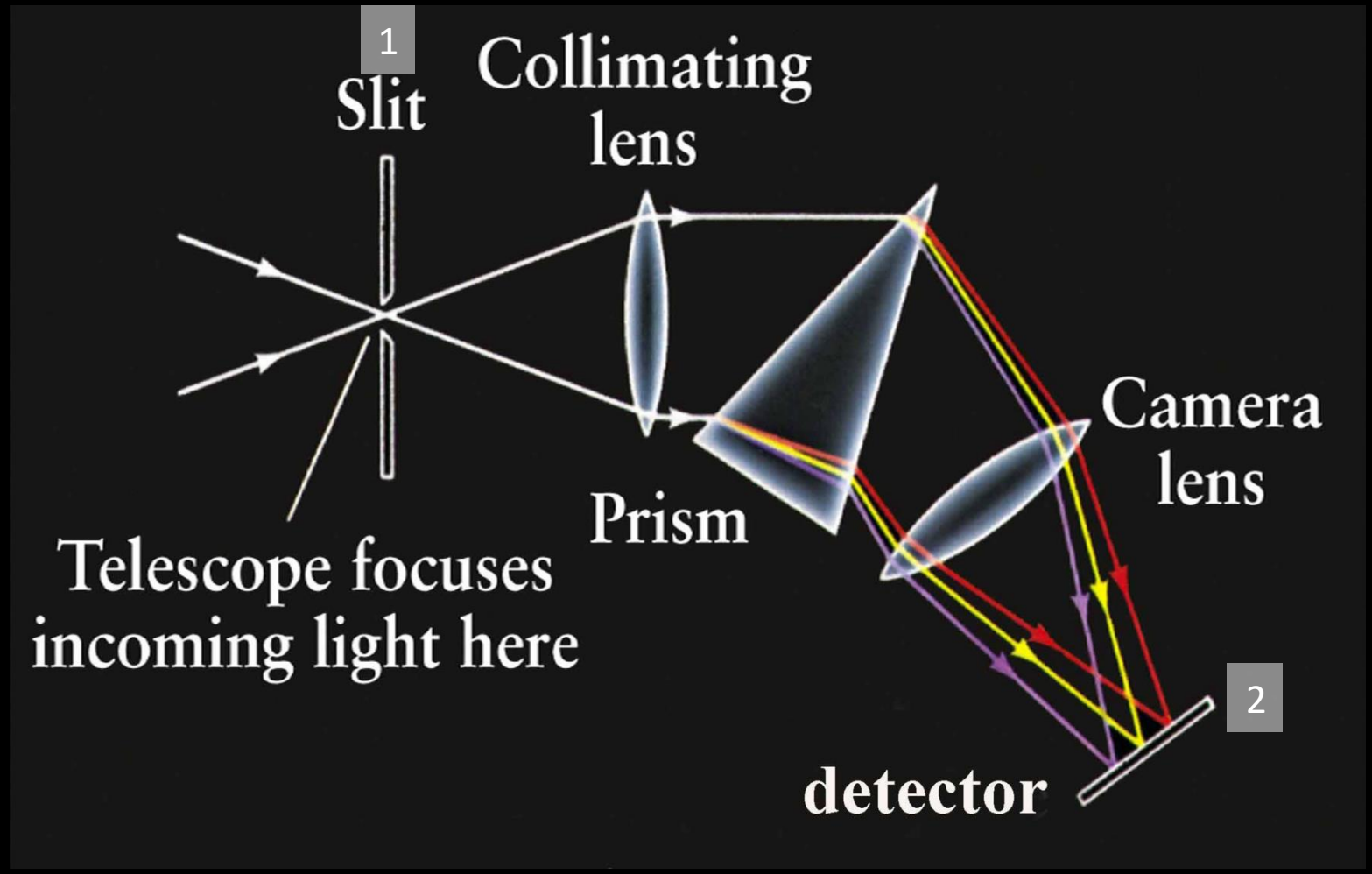
JWST/NIRSpec Grating Wheel



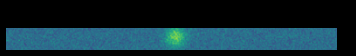


JWST/NIRSpec Grating Wheel (JWST Flickr Account/ESA). Blazed gratings (Gold) and a CaF2 prism (clear). These reflective gratings are optimized to maximize throughput.

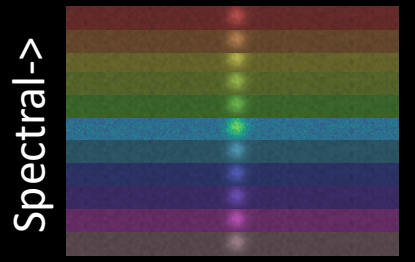
Impact of Components on Performance



Focal Plane w/ Slit 1

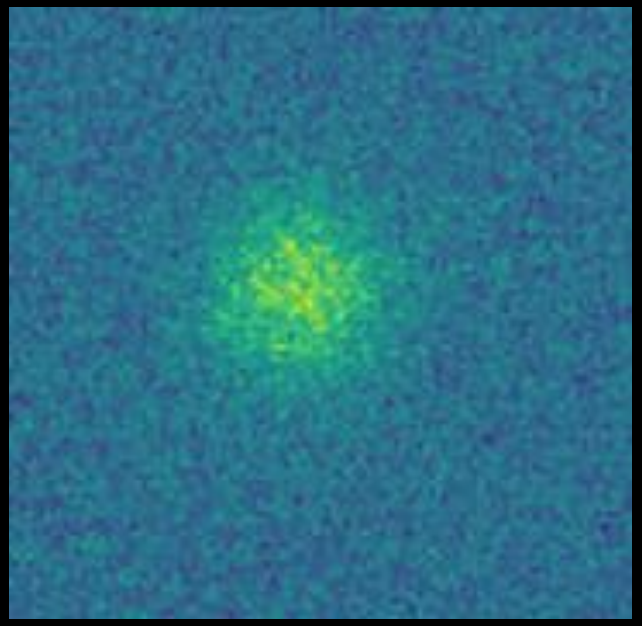


Detector 2

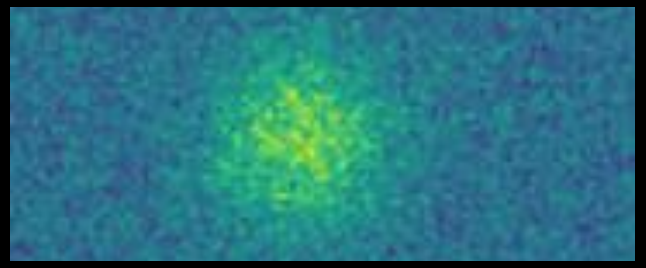


Spectrograph Slit Size Impacts Target Signal

Focal Plane no Slit



Focal Plane w Slit



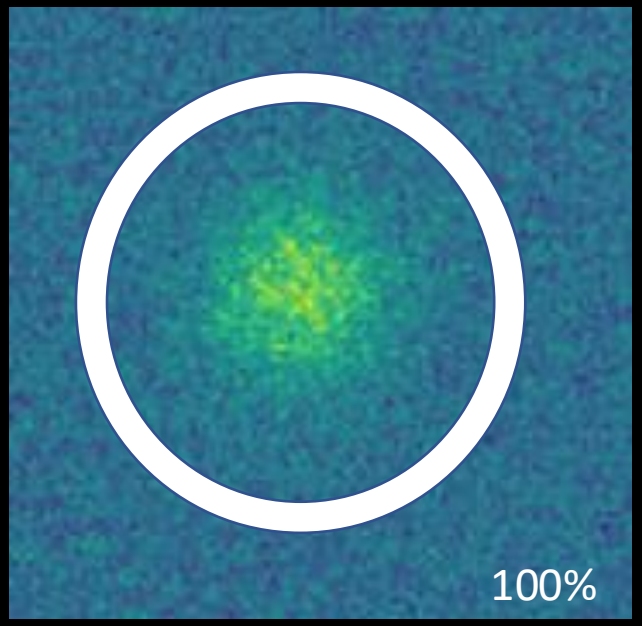
Focal Plane w Narrow Slit



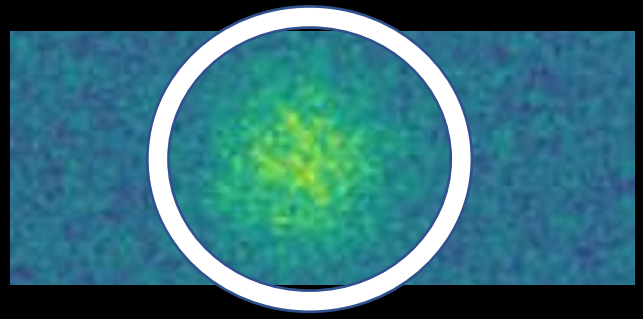
The size of a spectrograph's slit directly impacts the amount light you receive from a target. The aperture of the slit causes **slit losses** that can reduce the SNR of your observations.

Spectrograph Slit Size Impacts Target Signal

Focal Plane no Slit

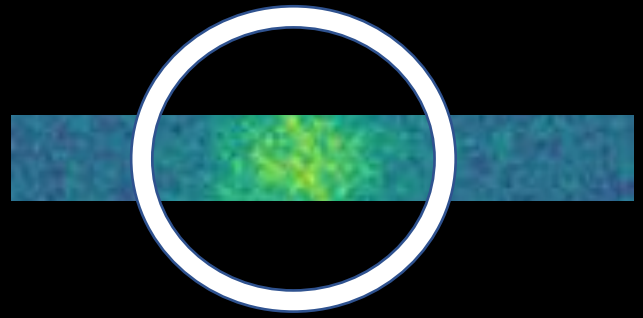


Focal Plane w Slit



95%

Focal Plane w Narrow Slit



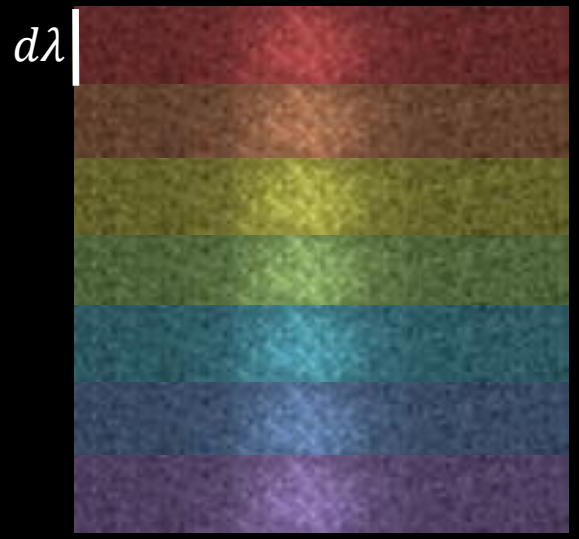
60%

If you cannot properly account for slit losses or have an undersized slit, you will not get an accurate photometric calibration.

Spectrograph Slit Size Impacts Resolution

Focal Plane w Narrow Slit





We are assuming some dispersive element with a fixed dispersion power ($d\theta/d\lambda$)

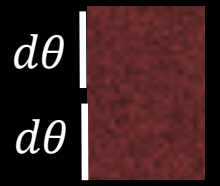
Spectrograph Slit Size Impacts Resolution

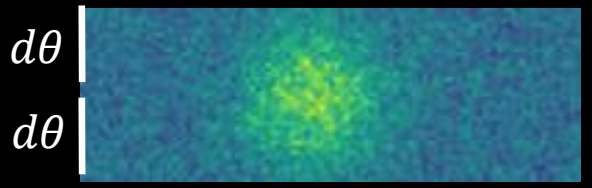
Focal Plane w Narrow Slit





Focal Plane w Slit



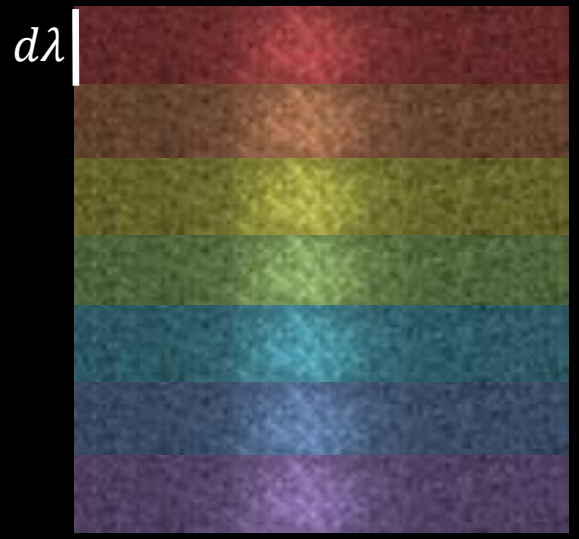


We are assuming some dispersive element with a fixed dispersion power ($d\theta/d\lambda$)

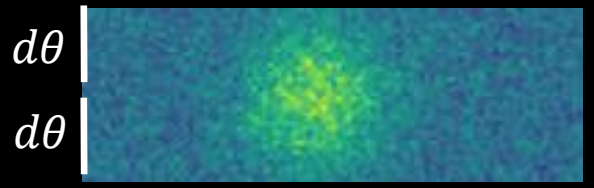
Spectrograph Slit Size Impacts Resolution

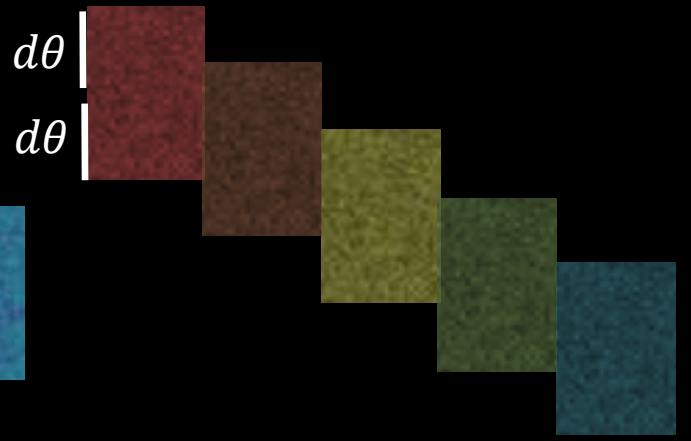
Focal Plane w Narrow Slit





Focal Plane w Slit



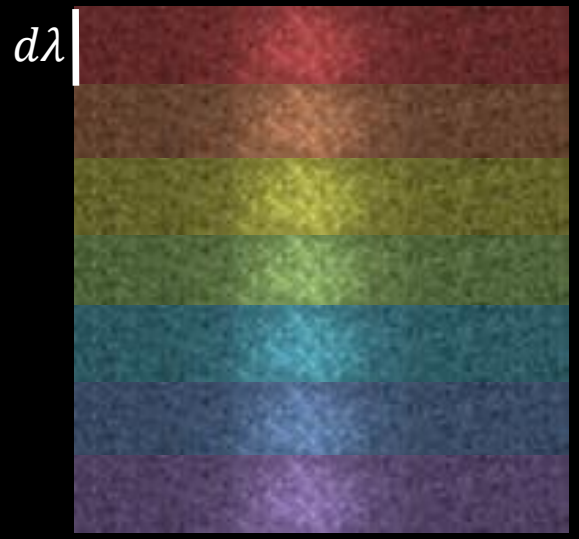


We are assuming some dispersive element with a fixed dispersion power ($d\theta/d\lambda$)

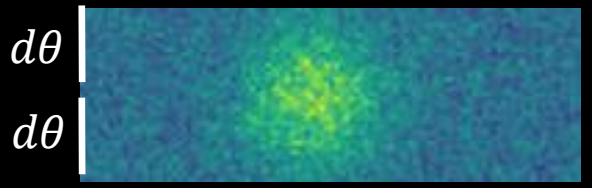
Spectrograph Slit Size Impacts Resolution

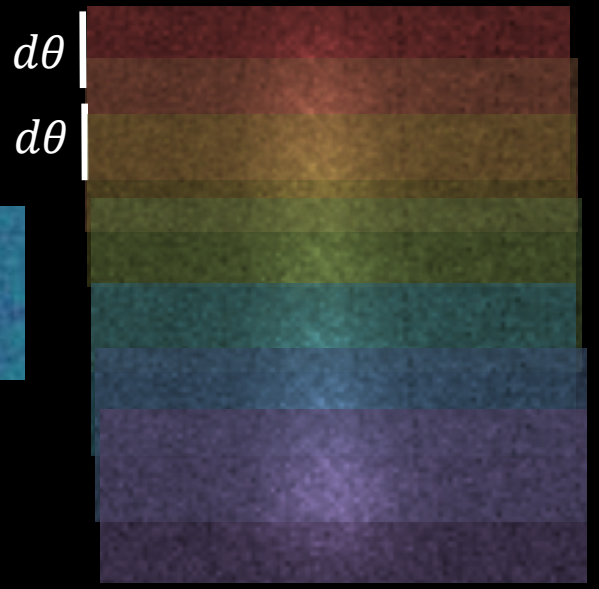
Focal Plane w Narrow Slit





Focal Plane w Slit



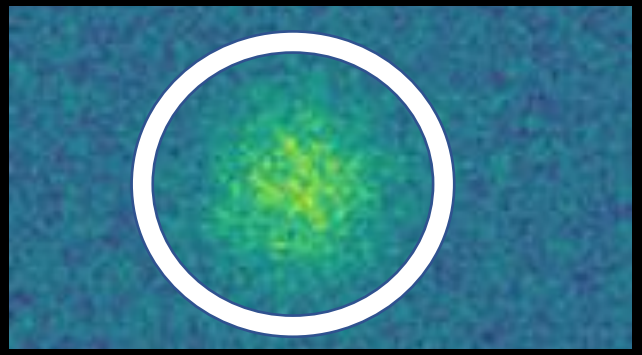


If you widen the slit, but keep the dispersion power the same your spectral resolution goes down. Nearby wavelengths start to overlap with each other.

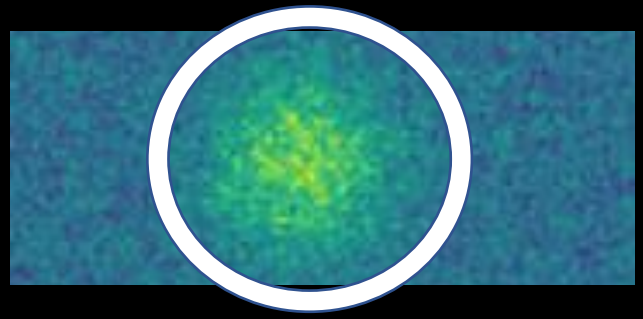
R ~ 1/w (slit width)

Choosing Slit Width Depends on Science Goals

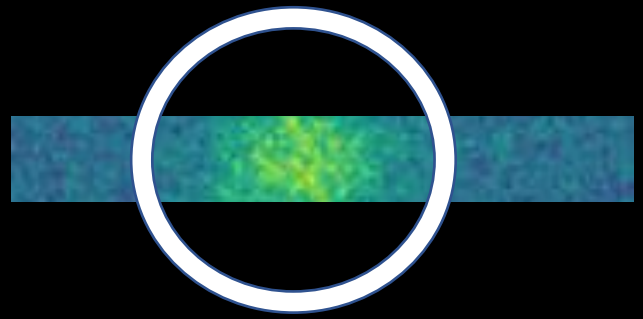
Focal Plane w Wide Slit



Focal Plane w Slit "Seeing/Diffraction Limit"



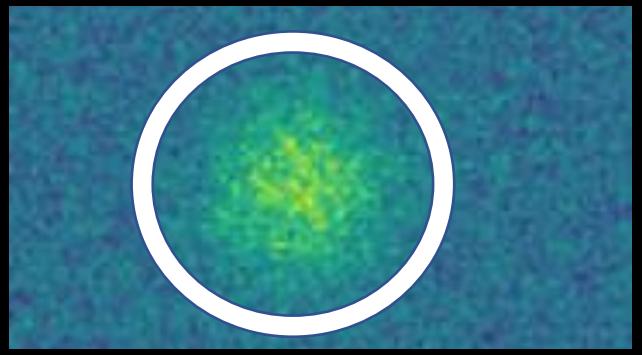
Focal Plane w Narrow Slit



- Good photometric stability
- Excess Sky or Background
- Low resolution (R~100)

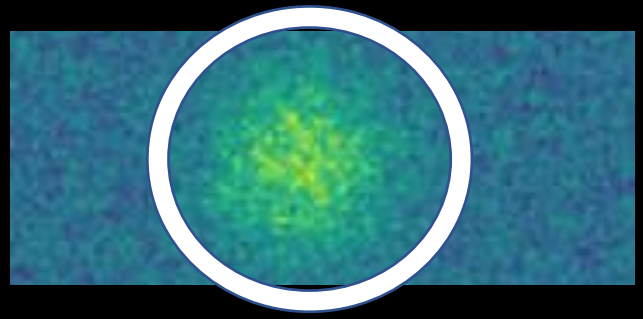
Choosing Slit Width Depends on Science Goals

Focal Plane w Wide Slit



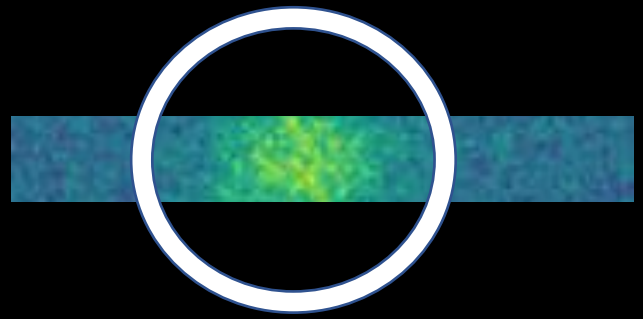
- Good photometric stability
- Excess Sky or Background
- Low resolution (R~100)

Focal Plane w Slit "Seeing/Diffraction Limit"



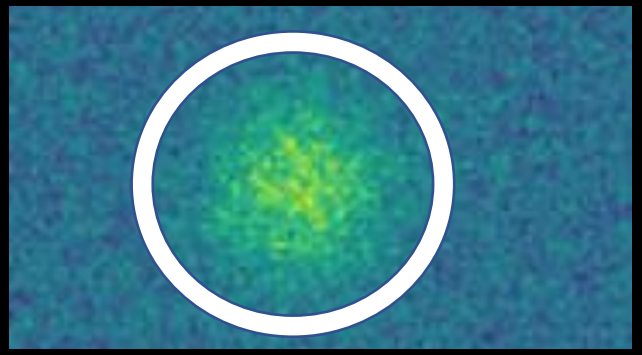
- Some photometric stability
- Medium resolution (R>1000)

Focal Plane w Narrow Slit



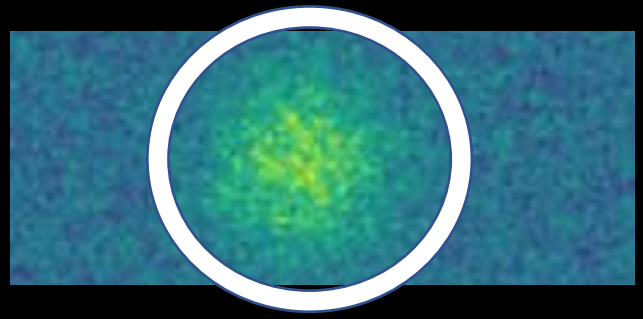
Choosing Slit Width Depends on Science Goals

Focal Plane w Wide Slit



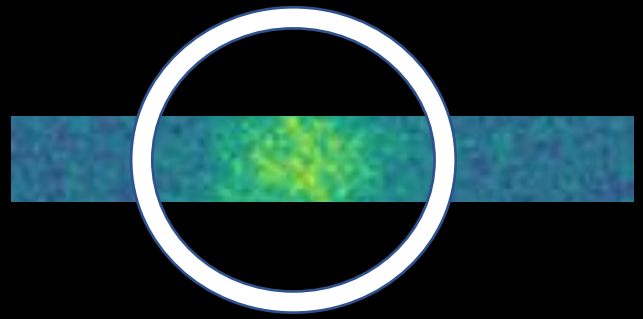
- Good photometric stability
- Excess Sky or Background
- Low resolution (R~100)

Focal Plane w Slit "Seeing/Diffraction Limit"



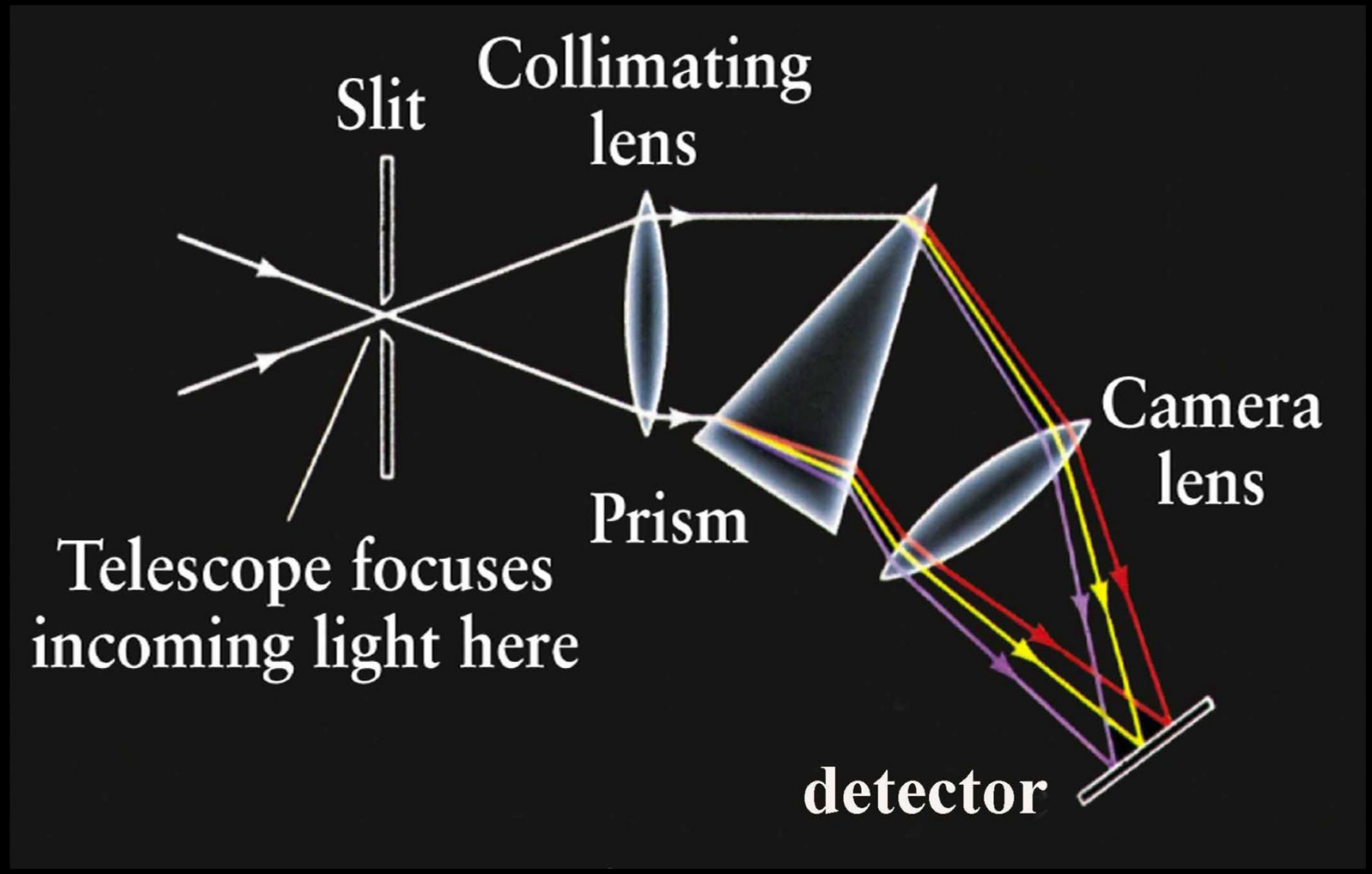
- Some photometric stability
- Medium resolution (R>1000)

Focal Plane w Narrow Slit



- Little to no photometric stability
- High res spectroscopy achievable (R > 10,000)
- Low signal to noise

Impact of Components on Performance



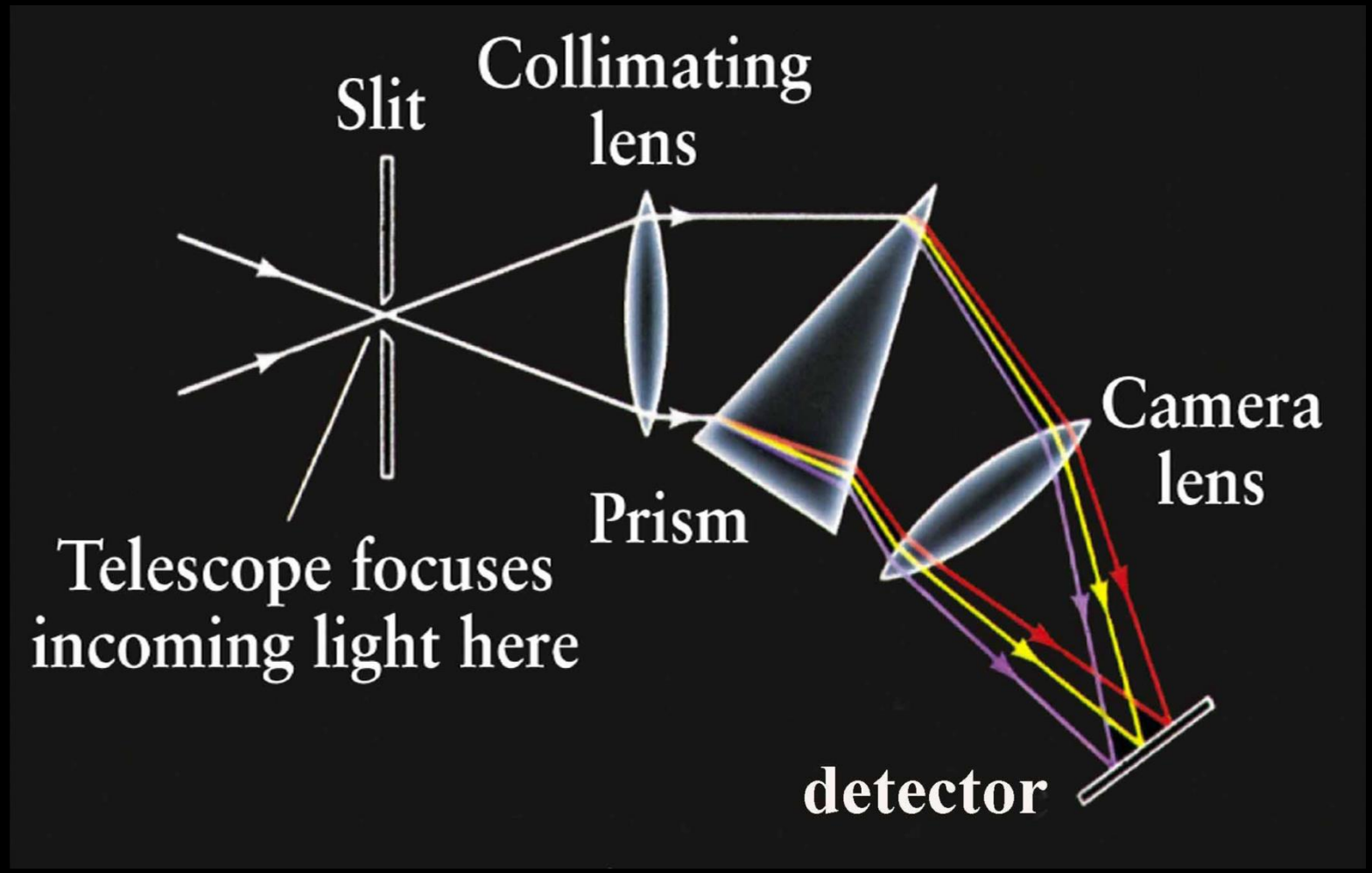
Collimator Lens

The collimating lens(s) does not directly impact spectral resolution.

The collimator however should not introduce any aberrations and should be **achromatic.**

Achromatic lenses don't break up the combined white light into components.

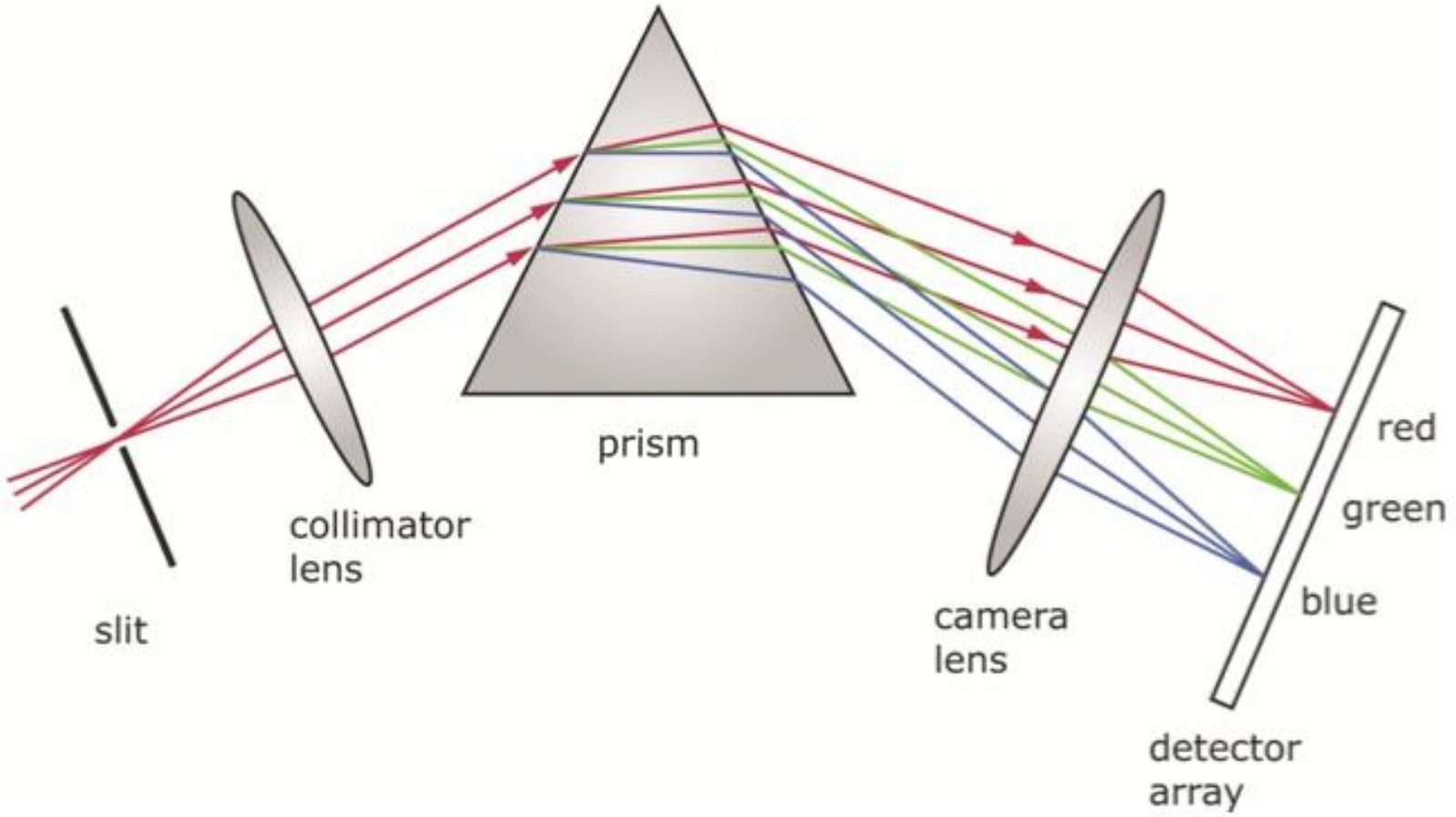
Impact of Components on Performance



Camera Lens

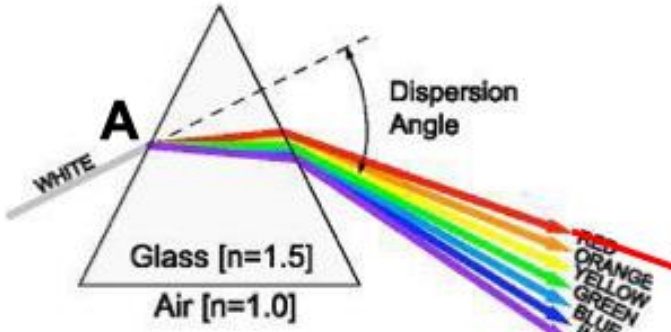
The size and focal length of the **camera lens(s)** impacts the spectral resolution and coverage of the spectrograph.

Impact of Components on Performance: Camera Lens Size



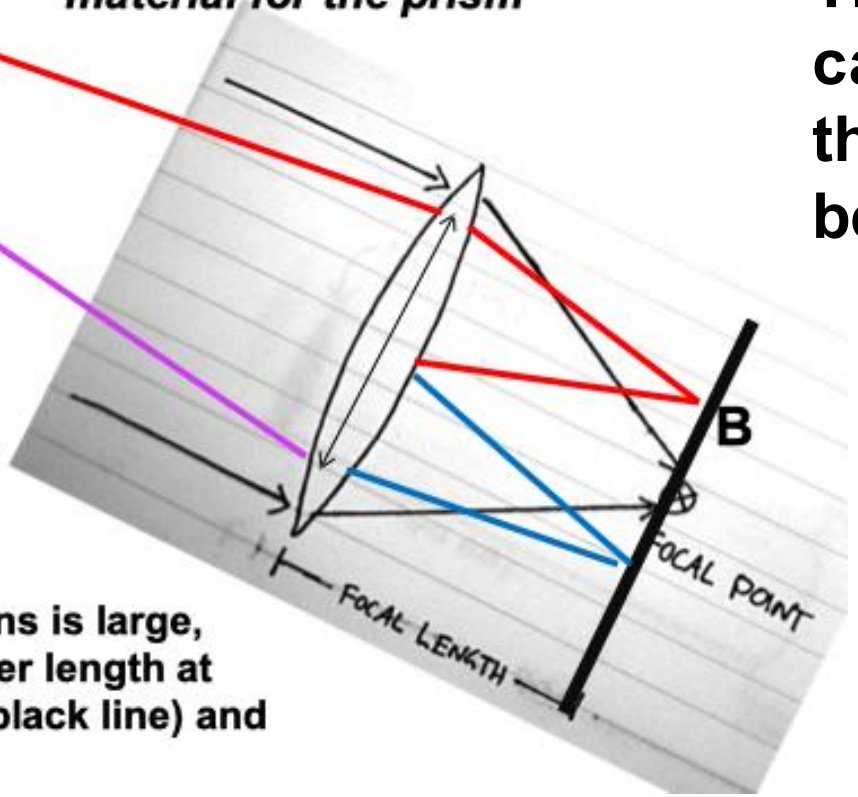
You need a camera lens large enough to capture the full angular extent of the spectrum you want. This sets the spectral grasp.

Impact of Components on Performance: Camera Lens Focal Length



We can increase the used diameter by adjusting the optical design or by selecting a more dispersive material for the prism

So the larger the used diameter of the camera lens, the higher the spectral resolution

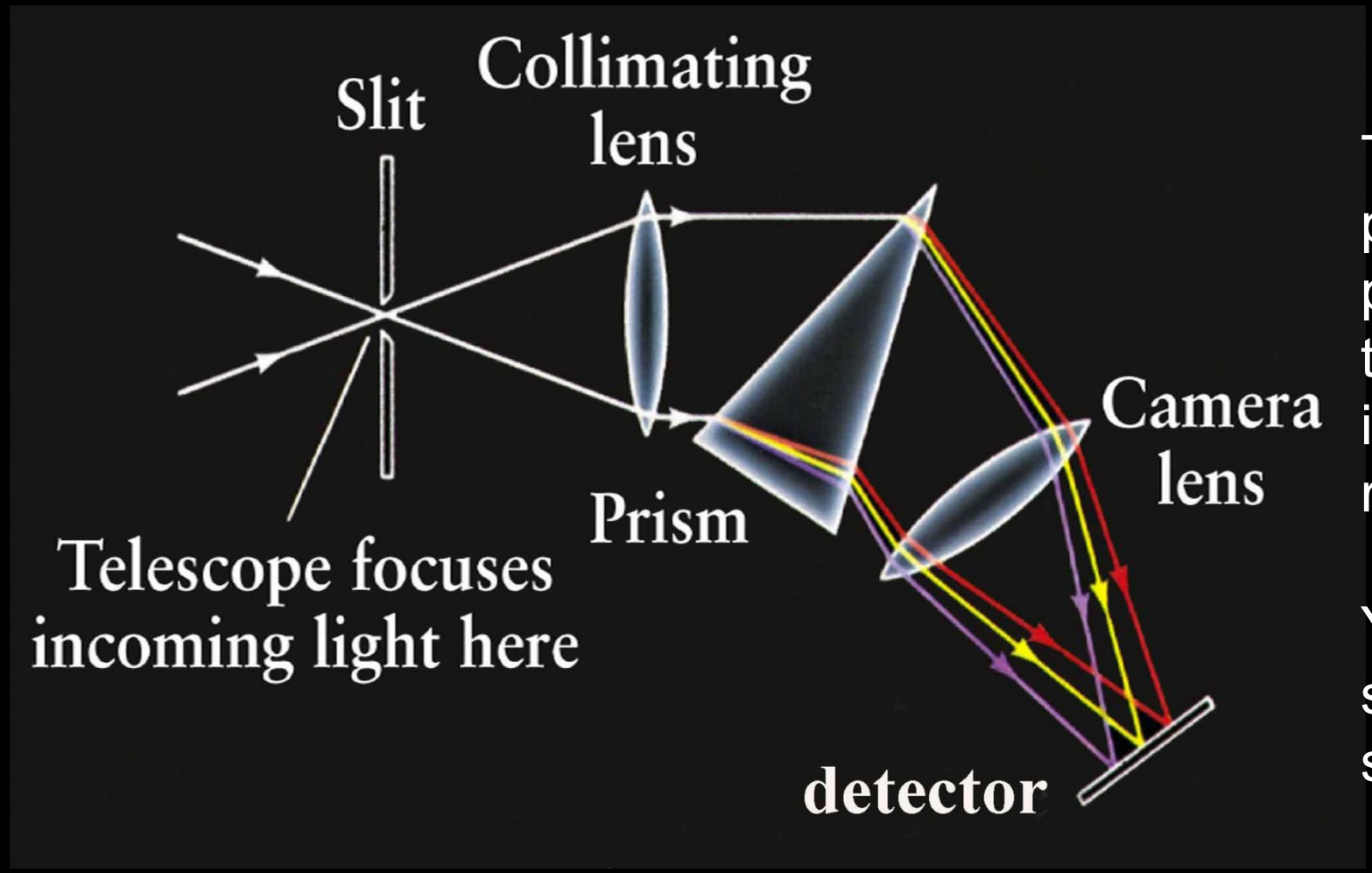


If the used diameter on the camera lens is large, then the spectrum is imaged to a larger length at the focal plane of the camera (heavy black line) and the spectral resolution is higher.

The focal length of the camera lens sets the angle of the focused beam again before the detector.

Slide from Peter Milne

Impact of Components on Performance



The CCD/Detector

The physical size of your pixels and the number of pixels you have is the last thing that determines the instrument's spectral resolution.

You cannot save more spatial information than the size of your pixel.

You often want 2 pixels per wavelength element.